Printed Pages: 2

Sub Code:KNC-402

Paper Id: 238061

Roll No.

B.Tech.
(SEM IV) THEORY EXAMINATION 2022-23
PYTHON PROGRAMMING

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

a. Define the programming cycle for Python

b. Describe the use of class `__init__()` method.

c. Explain features of any two Python IDEs.

d. Discuss the purpose of the break and continue statement in loops.

e. Explain mutable sequences in Python.

f. Explain the output of following function.

```
defprintalpha (abc_list, num_list):
    for char in abc_list:
        for num in num_list:
            print(char, num)
    return
printalpha (['a', 'b', 'c'], [1, 2, 3])
```

g. Discuss the special methods __le__ and __ne__.

h. Discuss various file opening modes of python.

i. Compare assert, try-except and raise statements.

j. Write python function to perform linear search.

SECTION B

2. Attempt any *three* of the following: **10x3=30**

a. Write Python code snippet to display n terms of Fibonacci series using recursion.

b. Write and explain an algorithm through python code to generate prime numbers.

c. Compare list and tuple data structure with suitable examples. Explain the concept of list comprehension.

d. Explain the python implementation of stack ADT.

e. Describe python program to implement selection sort. Simulate your code on the following data
2,52,1.5,2.54,62,64.

Source: AKTU official examination paper - www.aktunotes.in

SECTION C

3. Attempt any *one* part of the following: **10x1=10**

a. Discuss various categories of operators in python. Find and explain stepwise solution of following expressions if a=3, b=5, c=10.

i) a &b<<2//5**2+c^b
ii) b>>a**2<<2 >> b**2^c**3

b. Discuss why python is interpreted language. Explain history and features of python while comparing python version 2 and 3.

4. Attempt any *one* part of the following: **10x1=10**

a. Write a python program to count the vowels present in given input string. Explain the output of program through example.

b. Write a program that accepts sequence of lines as input and prints the lines after making all characters in the sentence capitalized.

e.g. If Input:
Hello world
Practice makes perfect
Then, Output:
HELLO WORLD
PRACTICE MAKES PERFECT

5. Attempt any *one* part of the following: **10x1=10**

a. Discuss list data structure of python. Explain various inbuilt methods of list with suitable example of each.

b. Compare list and dictionary data structure. Explain various dictionary methods with suitable examples of each.

6. Attempt any *one* part of the following: **10x1=10**

a. Discuss the concept of Iterators in python. Explain, how we can create text file in python? Describe python program to write the number of letters and digits in given input string into a File object.

b. Illustrate through examples, how we can implement different type of inheritance in python? Explain the static, instance and class method through suitable example.

7. Attempt any *one* part of the following: **10x1=10**

a. Explain the Tower of Hanoi problem and its recursive solution in Python.

b. Write python program to implement merge sort algorithm. Discuss, how we can compare the time complexity of two algorithms?

PAPER ID-411076

Roll No: | | | | | | | | | | | | | | |

BTECH
(SEM IV) THEORY EXAMINATION 2023-24
PYTHON PROGRAMMING

TIME: 3 HRS **M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data, then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 7 = 14**

a.	Give difference between = = and is operator.
b.	How we print the character of a given ASCII value in Python?
c.	Can you use else with a for loop? If so, when is it executed?
d.	What will be the output of the following Python code? l= [1, 0, 0, 2, ' hi', '', []] print(list(filter(bool, l))
e.	Describe the purpose of the split() method in string manipulation.
f.	What does the readline() function return when it reaches the end of a file?
g.	Which function is used to create identity matrix in NumPy?

SECTION B

2. Attempt any *three* of the following: **7 x 3 = 21**

a.	Explain the concept of dynamic typing in Python with an example.
b.	Explain how to define a list in Python. Write a Python program to remove duplicates from a list and print the resulting list.
c.	Explain the concept of functions in Python. Write a function that takes a list of numbers and returns the sum of all the numbers in the list.
d.	Write a Python program to read a file named "ABC.txt" and count the number of lines, words, and characters in the file.
e.	Explain the basic usage of matplotlib for plotting graphs. Write a Python program to plot a simple line graph showing the relationship between x = [1, 2, 3, 4, 5] and y = [1, 4, 9, 16, 25].

SECTION C

3. Attempt any *one* part of the following: **7 x 1 = 7**

(a)	Explain for and while loops used in Python with appropriate example.
(b)	Write a Python Program to find the LCM of two numbers.

4. Attempt any *one* part of the following: **7 x 1 = 7**

(a)	Write a program takes two strings and checks common letters in both the strings. Enter first string: Hari Enter second string: Hale The common letters are: H a
(b)	Write a Python Program to find the sum all the items in a dictionary. For example if d={'A':100,'B':540,'C':239}then output should be 879.

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BTECH
(SEM IV) THEORY EXAMINATION 2023-24
PYTHON PROGRAMMING

TIME: 3 HRS **M.MARKS: 70**

5. **Attempt any *one* part of the following:** **7 x 1 = 7**

	(a)	Explain the concept of a set in Python and its characteristics. How elements are added or removed in set.
	(b)	Explain how lambda functions can be used within a list comprehension. Write a Python program that uses a lambda function within a list comprehension to convert a list of temperatures in Celsius to Fahrenheit.

6. **Attempt any *one* part of the following:** **7 x 1 = 7**

	(a)	Explain different file opening modes also write a Python Program to read a file and capitalize the first letter of every word in the file.
	(b)	What do you mean by generators in Python? How it is created in Python?

7. **Attempt any *one* part of the following:** **7 x 1 = 7**

	(a)	Describe how to generate random numbers using NumPy. Write a Python program to create an array of 5 random integers between 10 and 50.
	(b)	Explain the concept of DataFrame in pandas. Write a Python program to create a DataFrame from a dictionary and print it.

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